BOLLAPALLY BANSI DEEPAK

Full Stack AI Engineer

+91-7330056795 | bansideepak2000@gmail.com | Hyderabad, India | BansiDeepak

PROFESSIONAL SUMMARY

Full Stack AI Engineer with **2 years of hands-on experience** building production web applications end-to-end: **React/Next.js** frontends, **Node.js/TypeScript** and **Python/FastAPI** backends, and **PostgreSQL** data layers. Writes tests first (Jest, Vitest, Playwright) and treats untested code as unfinished. Modernized a legacy **.NET** system into a fully tested TypeScript platform, and ships **AI features** that survive production: RAG pipelines, multi-model routing, and guardrails built on LangChain and the OpenAI/Gemini SDKs.

PROFESSIONAL EXPERIENCE

Full Stack AI Engineer

Neurom Innovations

Dec 2024 – Present | Hyderabad, India

- Build web applications end-to-end, from React/Next.js frontends through Node.js/Express and FastAPI services to PostgreSQL data layers, turning business requirements into shipped features alongside senior developers and architects.
- Own quality **test-first**: unit, functional, end-to-end and regression suites (**Jest**, **Vitest**, **Playwright**, Supertest) wired into GitHub **CI/CD** so defects surface before release, with TDD on high-risk business logic.
- Ship **AI features** into real products: LLM-backed workflows, retrieval agents, and schema-aware **RAG** pipelines built with **LangChain** and the OpenAI/Gemini SDKs, with guardrails (restricted tool-calling, structured output) that keep hallucinations out of user-facing answers.
- Built a multi-model inference gateway in **FastAPI** that routes each request between Gemini and OpenAI models based on task complexity, latency budget, and cost.
- Use AI coding tools daily (**Claude Code**, **GitHub Copilot**) for scaffolding, refactoring, test generation and code review, build custom **MCP** servers that give those tools project context, and recommend new AI tooling where the gain is measurable.
- Work the full **Agile** cycle, from sprint planning and story writing through regression testing, UAT and demos, and share a support rotation covering incident triage, root-cause analysis, and hotfix delivery.
- Keep long-running services production-healthy: dependency and framework upgrades, containerized

KEY PROJECTS

GROUP-BOOKING & TRAVEL MANAGEMENT PLATFORM – Legacy .NET Modernization

(In Progress)

TypeScript, NestJS, PostgreSQL/Prisma, Next.js, Jest, Vitest, Playwright, Docker

- Reverse-engineered an undocumented legacy **.NET** travel-management system (~32K lines C#, ~9K lines T-SQL) into a documented business-rules catalog, then rebuilt it as an API-first **TypeScript** platform: a **NestJS** modular monolith over **PostgreSQL/Prisma** with a **Next.js** staff admin, **110 REST endpoints**, and a **37-model schema**.
- Wrote **601 automated tests** across every layer: **Jest** for the API, **Vitest** for domain logic and web, **Playwright** for end-to-end in a real browser, with the pricing core built **test-first**.
- Designed pricing so a committed booking can **never be silently re-priced**: each booking pins the exact versioned price sheet it was quoted from, and all money lives in integer minor units so rounding can never drift a total.
- Prevented **overselling** of pre-purchased room inventory: database row locks serialize concurrent bookings before the capacity re-check, proven by a concurrency test that races real transactions against **PostgreSQL**.

CORAX – AI Face-Recognition Attendance Platform

FastAPI, PostgreSQL/pgvector, Celery, InsightFace, MediaPipe, React, React Native, Docker

- Architected a multi-tenant attendance platform end-to-end: an async **FastAPI** backend (**66 endpoints**), a **React + TypeScript** dashboard, and a **React Native** app sharing one typed package, shipped as a six-service **Docker** stack.
- Re-encrypted live **biometric data with zero downtime**: every encrypted record carries its own key version, so the system reads old and new formats side by side while records migrate.
- Closed **50 findings** across two security audits, including two IDOR classes and a login timing leak, and cryptographically bound each encrypted face image to its owner (**AES-256-GCM**).
- Ran a five-layer, **207-case** test plan across API, integration, security, web and mobile that surfaced **13 defects**, each traced back to its case ID.

Docker/Kubernetes deploys, and health and observability checks.

TECHNICAL SKILLS

Node.js | TypeScript | JavaScript | NestJS | Express | REST APIs | FastAPI | Python | Celery | Redis | nginx

Jest | Vitest | Test-Driven Development | Playwright | Supertest | E2E & Regression Testing

React.js | Next.js | React Native | Tailwind CSS | React Query

PostgreSQL | Prisma | SQL | pgvector | Docker | Docker Compose | Kubernetes | CI/CD | Git / GitHub | Agile / Scrum | AWS EC2

LLMs | RAG Pipelines | LangChain | OpenAI API | Google Gemini | AI Agents | Prompt Engineering | Vector Search | Text-to-SQL

AI Coding Assistants | Claude Code | OpenCode | GitHub Copilot | MCP

JWT | AES-256 Encryption | Multi-tenant Isolation | Rate Limiting

ASKENCE – Enterprise Text-to-SQL & Analytics Platform

React 19, TypeScript, FastAPI, LangChain, Gemini, PostgreSQL, Docker

- Built an enterprise Text-to-SQL platform on **LangChain**, choosing schema grounding over **RAG** retrieval after evaluating both: the full schema fits in the model's context, so retrieval would only add latency and a new way to fail.
- Collapsed **23 legacy region tables**, ten of them silently double-counting totals, into one denormalised fact table, because every JOIN a model must generate is one more chance to get the query wrong.
- Blocked a class of **silently-wrong answers** (numbers that are mathematically invalid yet still render as a clean chart) with three independent guards: a prompt rule, a non-additive column detector, and a shared display classifier.

CERTIFICATIONS

1. Oracle Cloud Infrastructure 2023 Certified Associate
2. Postman API Fundamentals Student Expert

EDUCATION

B.Tech, Computer Science & Engineering

Talla Padmavathi College of Engineering

2021 – 2025